Layout Validation Rules

How the 36 checks in the configurator's layout validator came together — the order they arrived in, what each one answers, and what earlier versions got wrong.

Rules	36, told as ten passages, plus a reference appendix
Tiers	HARD SOFT NOTE
Written from	`src/core/rules.ts`, `src/core/windows.ts`, `src/core/orientation.ts`
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1 · The tool and the graph

You place rooms, circulation, outdoor space and stairs on a 0.6 m grid across as many floors as you want, then put entrances on exterior edges and doors on interior ones. Pressing Check Layout runs all 36 rules and shows what they found in a written panel, a bubble diagram, and coloured tints on the shells in the 3D view. The checks are advisory: they run when you ask for them, and they never stop you placing anything.

Every rule reads one graph built from the whole dwelling. Its nodes are rooms, merged circulation and outdoor clusters, and stairs. Two spaces can be joined by a touch edge, meaning they share a wall, or an access edge, meaning a door connects them — a distinction the next section is really about, because most of what follows depends on it. Entrances sit on ground-floor exterior edges and are re-derived on every run, so building a room against an old entrance invalidates it.

What follows is roughly the order the rules arrived in. Most of them answer a question the previous one raised, or correct something the previous one got wrong. Each carries one of three tiers: hard, meaning the dwelling fails on it; soft, meaning the plan is unusual but often works fine anyway; or a note, meaning the tool recognised a pattern and is telling you it saw it, without judging. I'll say more about each tier as the rules that carry it come up.

2 · Entrances

Before anything else works you need a door to the outside. **E1** is the mildest rule in the set: if no entrance exists yet, it says so once, in place of every reachability flag that would otherwise fire with nothing to measure from. If you placed one but it's since been blocked, the message says that instead: something built across the entrance's edge, which needs a different fix than placing one from scratch.

E2 is what fires when a single entrance gets blocked this way. An entrance binds to one exterior cell edge; put a room or a cluster against that edge and it now faces an interior space, so nobody can come in through it. Entrance validity is re-derived from the exterior-edge geometry on every run, so a room built long after the entrance still catches it.

3 · Reachability: what "connected" means

The first real question a plan raises is whether you can get everywhere in it. **H1** is the direct check: is there a path of doors, including doors onto stairs, from an entrance to this room? If not, the room is orphaned — nobody can walk into it, so it fails, hard.

That rule only means what it says if "connected" means something real. My first version of the graph counted two rooms as connected whenever they shared a wall, which isn't the same claim. A wall between two rooms means they happen to be next to each other; it doesn't mean you can walk between them. So I moved reachability off shared walls and onto doors: a touch edge in the graph now just records that two spaces share a wall, and only an access edge — an authored door joining them — counts for anything **H1** or its relatives compute. Circulation stopped being a byproduct of the floor plan's geometry and became something you state on purpose, one door at a time.

That change has a visible cost on a plan you're still building. Place a dozen rooms with no doors between any of them and **H1** flags every one, because none of them are connected to anything yet. **DR1** exists to explain that wall of red so it doesn't read as a bug. It's a note — the lightest of the three tiers, meaning the tool isn't judging anything, just telling you what it saw — and it says plainly what's going on: no doors placed yet, so nothing is reachable yet. It goes quiet the moment you place your first door.

Stairs get the identical treatment. **ST1** wants a door at both ends of a stair — one on the floor it leaves, one on the floor it arrives at, the second reached through the projection of the stairwell hole onto the floor above — and names whichever end is missing. **ST2** is **ST1**'s harder cousin, and it's really **H1** for vertical circulation: no doored route from any entrance reaches the stair, so the floor it serves is cut off along with it.

I also tried, early on, requiring a dedicated corridor between any two connected rooms. That rule doesn't exist anymore. A door straight from a living room into a bedroom is as legitimate a route as one that passes through a hallway, and plenty of working flats are built that way. Reachability only checks whether a path of doors exists between two rooms.

4 · Reachable, but reachable how

Once "reachable" meant something, a sharper question followed: reachable by what route? A room can have a perfectly good doored path to the entrance and still be badly placed, if that path happens to run through somebody's bedroom or through the one bathroom in the flat.

Three rules ask this the same way. Run the breadth-first search from the entrances once, normally, to get the full reachable set. Run it again with one room type barred from being crossed — you can still enter a room of that type, since it might be your destination, but you can't walk through it to get somewhere else. Anything present in the first set and missing from the second can only be reached by crossing the barred type.

> A small example. The entrance is on the living room. A door leads to a bathroom, and a door from that bathroom leads to a bedroom with no other door of its own. The full search reaches all three rooms. Run it again with bathrooms barred as through-routes, and it reaches the living room and stops there — the bathroom is off limits as a route now, even though you could still walk into it as a destination. The bedroom is in the first set and missing from the second, so **H2** flags it: the only way in is through the bathroom. Add a second door straight from the living room to the bedroom, and the barred search now reaches it too, and **H2** goes quiet.

H3 runs the same search with bedrooms barred, and it's the more consequential of the two — crossing someone's bedroom to get anywhere is a privacy failure, whether the room behind it is another bedroom or a stair. It caught something I hadn't intended, though: any bathroom reachable only through its own bedroom got hard-flagged. That isn't a mistake in the plan — it's what an en-suite looks like in this graph. So **H3** exempts bathrooms from the flag. What used to be a false failure became **S7**, a note that just names the pattern: this bathroom is reached through its bedroom, and that's fine. **H3** still fires for every other room type, and it still fires for stairs — vertical circulation gated behind a bedroom door is the kind of thing this rule exists to catch.

Exempting bathrooms from **H3** leaves an actual question unanswered: can a guest reach a toilet at all without crossing someone's private room? That's a dwelling-wide question, so it's a separate rule: **G1**, and it's the first soft rule in this account. A rule is soft when the plan has departed from ordinary practice without being unusable, and small flats built around one en-suite bathroom are common enough that hard would overstate the problem. **G1** runs the bedroom-barred search once for the whole dwelling and fires if no bathroom survives it.

There's a second gap the barred searches leave, and it comes from how they treat their own starting point. You always enter through the room hosting the entrance — that's what walking in the front door means — so the search never bars the entrance's own host, even if it's a bedroom. A studio whose front door opens straight into its only bedroom never trips **H3**, which is correct, but it's still worth knowing about, so **G2** flags it directly: this entrance opens into a private room. Soft, because studios do this on purpose.

H6 is the outdoor case of the same pattern — a room reachable only by crossing a balcony or terrace is at the mercy of the weather before anyone can use it — and it needs no extra guard the way **H2** and **H3** do: outdoor space is a cluster node, so it drops out of the room-and-stair target set on its own.

5 · A door in the wrong place

A kitchen and a bathroom sharing a wall is ordinary practice: it's how the plumbing stays cheap, with both rooms stacked on one riser and short pipe runs. My first version of **H4** read the shared wall and flagged that arrangement. That was backwards: the hazard is a door straight from food preparation into a toilet. The wall on its own is fine. So I moved **H4** onto access edges: it now fires only when a door joins a bathroom and a kitchen, and leaves the shared-wall case alone.

The wall case earned its own note instead, **S6**, which fires where **H4** doesn't — a kitchen and bathroom sharing a wall with no door through it — so the two never land on the same boundary. The House-GAN failure mode behind the earlier version described a closet built inside a kitchen: a containment problem, and it never really supported flagging a shared wall in the first place.

6 · Measuring depth

Space syntax has a way of measuring how buried a room is in a plan: count the doored steps from the entrance to get there. I compute that for every room, and **DP1** is the first thing built on it — flag any room five or more steps in. The threshold is `DEEP_ROOM_THRESHOLD_HOPS = 5`, and it's one of the numbers here I'd most like to be argued out of.

Getting that count right needed one correction. A stair is a graph node like any room, so a plain count charges two hops for a floor change — one to enter the stair, one to leave it — and every room on the floor above drifts toward the threshold just by existing there. I weight the two differently: entering a stair costs a step, leaving one costs nothing, so a floor change costs one hop overall, the same as any other doored step. That keeps the threshold meaning what it meant on a single floor once a second one is added.

Once every room has a depth, you can also ask about groups of rooms. Hillier and Hanson describe the depth pattern that recurs across a building type as its genotype: strip away the shape and what's left in common is which rooms sit shallow and which sit deep. For a dwelling the pattern holds fairly consistently — the rooms where you receive people sit near the entrance, and the rooms where you sleep sit further in, so a visitor is carried through the social part of the plan before reaching the private part. **PG1** tests for the inversion of that pattern: compare the mean depth of the public rooms, living and recreation, against the mean depth of the bedrooms, and flag if the public mean comes out larger — bedrooms shallower than the sitting room, so a guest reaches private space before social space. It needs at least one reachable room in each set, and it needs an entrance to measure from at all. Both means print in the report whether or not the rule fires, so the gradient is legible on the plans where **PG1** stays quiet too.

F1 reuses the same hop-counting machinery for a different question: not how buried is this room socially, but how far is it from a way out. It seeds the same kind of search at every entrance and every stair together, since a stair is the way off whatever floor you're standing on, and flags anything past `ESCAPE_DEPTH_MAX = 4` hops from the nearest one. It counts door-hops on the access graph. Real fire code measures metres of travel distance and wants a second independent escape route as well; neither of those is modelled here. **F1** approximates the idea with a hop count.

> Take an entrance on the ground-floor living room, a door to a hall, a door from the hall onto a stair, and upstairs a corridor with a chain of doors running through two rooms to a bedroom. Counted from the entrance: hall at 1, stair at 2, corridor at 2 as well, since leaving the stair costs nothing, then 3, then 4, then the bedroom at 5 — **DP1** flags it. Counted from the stair, which **F1** also treats as a seed: the corridor is 0, then 1, then 2, and the bedroom sits 3 hops from a way out, so **F1** says nothing about it. The bedroom is deep in the social sense and close to an exit at the same time; the two rules report each half of that.

7 · Circulation: corridors, balconies, width

A corridor earns its floor area by connecting spaces, so the first two checks on circulation ask how well it's doing that. **C1** flags a circulation cluster with no doors onto it at all — dead space, circulating nothing. **C2** flags one with a single door, which lets you step in and step back out the way you came but joins nothing to anything. Both read access degree, the count of doors onto the cluster. Before reachability moved onto doors, degree counted the rooms a corridor happened to touch through sealed walls, so a corridor could sit surrounded by rooms and register as well connected while having no way in at all — neither rule could have caught that.

C1 started out hard and isn't anymore. A dead corridor wastes plan area, but the flat around it still works, so hard overstated what the rule was finding. It's soft now, alongside **O1** — the same zero-degree condition applied to outdoor space instead of circulation, a balcony or terrace nothing opens onto, unusable for the same reason a doorless corridor is. The semi-exterior pass changed what "opens onto" means: a room's boundary with a balcony is a french window, and the glass IS the door, so no authored door is needed. O1 is now gated to the pre-entrance case and the hard **OD1** carries the real claim — a balcony you cannot reach from the entrance is unusable floor area. A 1-cell contact still confers nothing: 0.6 m of wall is not a way through.

S1 sits at the other end of the same axis: an outdoor space with more than two doors onto it usually means it's being used as a route between rooms, when the ordinary use is stepping out and back. **S2** asks a related question about the living room — one door or none, when the room best positioned to be the flat's social hub is acting like a dead end instead. Both thresholds trace to the connection counts in House-GAN's Table 1, which were measured on physical adjacency; this model counts doors, so the numbers are approximate against that data.

A well-connected corridor can still fail on a different axis: width. SIA 500 wants roughly 1.2 m of clear passage for a wheelchair, two cells on this grid, and nothing before **A1** looked at width at all — a plan could pass every other check with every corridor a single 600 mm cell wide, while the doors opening onto those corridors were 1200 mm.

Counting neighbours doesn't find a narrow corridor reliably. Take an L-shaped corridor one cell wide. The cell at the corner has a neighbour to the west and a neighbour to the south, so by a neighbour count it looks the same as a cell in the middle of a proper two-wide hall, which also has neighbours on two axes. The corner is still a 600 mm passage. So **A1** asks about area instead: a cell has accessible width if it belongs to at least one 2×2 block of cells lying wholly inside the same cluster. Four such blocks contain any given cell, and if none of them is fully occupied by that cluster, the cell is narrow. A one-wide corridor fails at every cell, corner included, because no 2×2 square fits anywhere along a one-wide run. A two-wide corridor passes everywhere. A wide hall with a one-wide spur off it flags only the spur. **A1** reports once per cluster, with the count of narrow cells in the message, and it only looks at circulation — rooms come from fixed presets, and outdoor space is out of scope.

8 · Adjacency by touch

A few checks read the shared wall itself, because the thing they're describing is a fact about adjacency. **S3** flags a bedroom sharing a wall with a kitchen, a living room or a recreation room — noise and privacy bleeding through a wall that a corridor or a service space could have absorbed instead.

I had an earlier rule in the same family, flagging two bedrooms sharing a wall, and it didn't survive testing. Two quiet rooms next to each other is how most flats are built, and I couldn't find anything to ground a rule against it. What replaced it, **AC1**, targets something groundable: a bedroom sharing a wall with a stair, which carries footfall as impact noise through the structure and airborne noise through the air, and SIA 181's separation of noisy uses from quiet ones speaks directly to that case. **AC1** is scoped to stairs alone — a bedroom against a recreation room already goes through **S3**, since recreation counts as a public room there, and firing both on the same wall would add nothing.

S5 is the door version of a similar adjacency, and it isn't a warning: a door between a kitchen and a living room reads as an open-plan layout, and the note confirms the tool saw the pattern. A kitchen and living room that only share a sealed wall get nothing from **S5**, since a sealed wall between them says nothing about how the plan is meant to work.

Two more notes just flag counts, without judging them. **P3** mentions when a dwelling has more than one kitchen — unusual for a single flat, common enough in a shared house or a granny annex. **DR2** mentions a bedroom with three or more doors, which starts to eat into both the wall space a bedroom needs for furniture and the privacy it's supposed to offer; most bedrooms carry one door, sometimes two if there's an en-suite behind one of them.

9 · Program

A few checks aren't about the graph's shape at all, just about what's present. **P1** and **P2** ask whether a bathroom and a kitchen exist anywhere in the dwelling; without either one the flat doesn't work, hard, and there isn't much more to say about them.

MB1 is a softer version of the same idea, scoped to one floor at a time: bedrooms present, no bathroom on that floor, which on a multi-storey flat means an inconvenient trip down the stairs at three in the morning, so it's soft. It only runs once a bathroom exists somewhere in the dwelling — a bathroom-less flat is already **P1**'s problem, and **MB1** is gated on **P1** being satisfied so the two don't both fire over the same missing room.

10 · Daylight and glazing

D1 asks whether a bedroom, living room or recreation room has any exterior wall at all — no facade means no daylight, and no interior arrangement fixes that, so it's hard. **D2** is the same question for kitchens, soft instead of hard, because an internal kitchen is common enough and gets mechanically ventilated as a matter of course.

D1 depended on getting "exterior wall" right, and my first version of that test didn't handle stairwells correctly. An edge facing the stairwell hole used to count as exterior, so a room on an upper floor bordering the opening passed **D1** with no real facade, and the window generator could put glass on an edge that faced straight down an open shaft. I fixed the edge classification to read the same occupancy map the door system already used, which understands a stairwell hole as occupied space, and **D1** now catches those rooms.

W1 goes a step further than "any exterior wall" and asks whether the room has enough glazing on the wall it has. Swiss practice targets roughly a tenth of floor area in glass for a habitable room; bedrooms and kitchens use that figure directly, `1/10`. I set living rooms and recreation rooms higher, at `1/6`, since those are the rooms occupied through the daylight hours in a way that supports a larger opening. A kitchen gets a fixed two-cell band instead of a ratio, since a kitchen window is really sized by the counter run beneath it.

> Work through a bedroom at the default floor height of 3.0 m to see how the numbers land. Twelve cells of floor area is 4.32 m^2, so the target is 0.432 m^2 of glass. A framed edge — the bedroom variant, with a sill and a lintel each taking 900 mm — supplies $0.6 \times (3.0 - 0.9 - 0.9) = 0.72$ m^2, so one edge already clears the target, and it's really the two-edge minimum that decides the window size: the room gets a 1.2 m band, close to a third of its own floor area in glass. The ratio only starts to govern room size on larger rooms. A full-height living-room edge supplies $0.6 \times (3.0 - 0.9) = 1.26$ m^2, so a living room needs more than about 15 m^2 of floor before a sixth of its area calls for a third edge.

W1 fires when the exterior wall a room actually has can't supply the edges it needs — in practice, when the facade is broken into runs shorter than two cells — and it only evaluates rooms that have some exterior wall to glaze. A room with none belongs to **D1** or **D2** instead.

A band that runs out of room on its own straight edge doesn't just stop and open a second, disconnected one elsewhere — first it tries to turn a convex corner onto the adjoining edge, continuing as an L in plan, or, if it can turn at both ends of the same run, a U. The corner has to be convex, two edges of the same cell facing outward, because that's the only shape a plan can turn that way; a notch is two different cells meeting from opposite directions, which the wrap never bridges — there's no separate check for that, the same-cell test simply can't see it. An entrance on the corner edge stops the wrap exactly as it stops a straight band, since the wrap only ever looks at edges the generator already treats as exterior. The glazing still meets without a corner post: the north/south pane already reaches the true corner — it owns that corner square, the same as a plain wall does — so it's the east/west pane that extends past its usual reveal to the same point, overlapping it by a hair at the seam rather than falling short. That asymmetry has a second effect: either pane, alone, still reads as a complete edge if the other is cut away by the camera, since neither one's extent depends on whether the other is drawn.

Which face the band lands on is now a question of orientation. The project carries a north direction — a free angle you set with a compass dial — defined once, precisely, as world −Z rotated clockwise (viewed from above) by that angle; everything that needs a bearing reads that one definition. The generator scores each candidate exterior run by how close it faces due south and takes the southernmost first, so glazing migrates to the sunny faces on its own: rotate north and the bands slide around to whatever wall now points south. **OR1** is the rule that watches the result — a habitable room or a kitchen whose every window ends up facing within 45° of due north is lit only from the north, no direct sun, and gets a soft flag. It reads the same derived glazing the generator produced, and its flag is set only when the room actually HAS glazing and all of it is northern, so a room with no windows at all stays **D1**'s or **W1**'s business and never doubles up here. The tier is soft on purpose: a north-only room is a real comfort compromise at this latitude, not a code failure, and some rooms are fine that way. The report also prints each room's glazing orientation as plain information — "Living Room: glazing S + E" — so the gradient is legible whether or not the rule fires; a corner-wrapped band naturally shows both of its sectors.

What follows is reference material: the full rule list for lookup, the numbers the report prints alongside the rules, where the model stops short of a fuller analysis, and the sources behind the rules above.

A · All 36 rules, for lookup

ID	WHAT IT CHECKS	TIER
A1	A circulation cluster contains a cell narrower than 1.2 m clear width.	SOFT
AC1	A bedroom shares a wall with a stair.	SOFT
C1	A circulation cluster has no doors onto it (orphaned corridor).	SOFT
C2	A circulation cluster is reached by a single door.	SOFT
D1	A habitable room (bedroom, living, recreation) has no exterior wall.	HARD
D2	A kitchen has no exterior wall.	SOFT
DP1	A room is 5 or more door-hops from the entrance.	SOFT
DR1	No doors are placed yet; explains the reachability flags this causes.	NOTE
DR2	A bedroom has three or more doors.	NOTE
E1	No working entrance exists yet to validate reachability from.	NOTE
E2	An entrance's edge no longer faces outside.	HARD
F1	A room is more than 4 door-hops from the nearest entrance or stair.	SOFT
G1	No bathroom is reachable without crossing a bedroom.	SOFT
G2	An entrance opens directly into a bedroom or bathroom.	SOFT
H1	A room has no doored path from any entrance (orphaned room).	HARD
H2	A room or stair is reachable from an entrance only by crossing a bathroom.	HARD
H3	A room or stair (bathrooms exempt) is reachable only by crossing a bedroom.	HARD
H4	A door joins a bathroom and a kitchen directly.	HARD
H6	A room or stair is reachable only by crossing outdoor space.	HARD
MB1	A floor has bedrooms and no bathroom (checked once a bathroom exists somewhere).	SOFT
N1	Circulation area exceeds 25% of the interior — whole-dwelling, and independently on any single floor of a multi-floor plan.	SOFT
O1	Nothing opens onto an outdoor cluster (pre-entrance only — OD1 owns it afterwards).	SOFT
OD1	An outdoor cluster is not reachable from the entrance.	HARD
OR1	A habitable room or kitchen is glazed only toward the north (all windows within 45° of due north).	SOFT
P1	No bathroom exists anywhere in the dwelling.	HARD
P2	No kitchen exists anywhere in the dwelling.	HARD
P3	More than one kitchen exists in the dwelling.	NOTE
PG1	The mean depth of public rooms exceeds the mean depth of bedrooms.	SOFT
S1	An outdoor cluster has more than two doors onto it.	SOFT
S2	A living room has one door or none.	SOFT

ID	WHAT IT CHECKS	TIER
S3	A bedroom shares a wall with a kitchen or a public room.	SOFT
S5	A door joins a kitchen and a living room (open plan).	NOTE
S6	A kitchen and bathroom share a wall with no door between them.	NOTE
S7	A bathroom is reachable from an entrance only through its bedroom (en-suite).	NOTE
ST1	A stair has no door on its bottom and/or top landing.	SOFT
ST2	A stair is not reachable by doors from any entrance.	HARD
W1	A room's achieved glazing falls short of its daylight target.	SOFT

B · Metrics the report prints

Four figures the report shows on every run, whether or not the rules reading them fire.

Entrance depth. For each reachable room, the number of doored steps from the nearest entrance. The report gives the maximum, the mean, and a per-room list; the diagram puts the number on the node. Stairs are weighted as in §6, so a floor change costs one hop overall. **DP1** reads this map, and **PG1** reads its group means.

Circulation percentage. Circulation-cluster cells plus stair cells, over all occupied interior cells, with outdoor excluded from both. Printed as "Circulation: N% of interior area", and, once the dwelling has more than one floor, a second line breaking the same figure out per floor ("Floor 0: N% · Floor 1: N%") — omitted on a single floor, where it would just repeat the line above. **N1** flags above 25% whole-dwelling, and separately flags any floor that crosses 25% on its own even when the average across floors doesn't; the two flags measure different things and either or both can fire on the same dwelling.

Glazing against target. Per room, the glazing area the generator managed to place, set against the target for that room's type. **W1** flags the shortfall.

Public and bedroom mean depth. The two means **PG1** compares, printed as "Public mean depth X · Bedroom mean depth Y".

C · Assumptions and limits

- **F1** counts door-hops on the access graph. Fire code measures metres of travel distance and asks for a second independent escape route; neither is modelled.
- The House-GAN connection counts behind **S1**, **S2** and **DR2** came from proximity adjacency; this model counts doors, so the thresholds are approximate against that data.
- There is now a compass — a free north angle you set — and glazing is biased toward south with **OR1** flagging north-only rooms. But that's the whole of the orientation model: one north/not-north cut at 45°, no seasonal sun angles, no per-sector solar-gain scoring, no shading from neighbours. Daylight beyond that stays a question of exterior wall and glazed area.
- Rooms come from fixed presets, so minimum areas and dimensions are built into the presets and never checked. **A1** is the one dimensional test in the ruleset, because circulation is the one thing the user

shapes freely.

- One dwelling, no site: no neighbours, no party walls, no structural stacking. An upper-floor cell cantilevered past the floor below passes without comment.
- The named constants — `DEEP_ROOM_THRESHOLD_HOPS = 5`, `ESCAPE_DEPTH_MAX = 4`, `CIRCULATION_FRACTION_MAX = 0.25`, **DR2**'s three doors, the glazing ratios — are starting points I chose. None of them are derived optima.

D · References

Nauata, N., Chang, K.-H., Cheng, C.-Y., Mori, G., & Furukawa, Y. (2020). *House-GAN: Relational Generative Adversarial Networks for Graph-constrained House Layout Generation*. European Conference on Computer Vision (ECCV) 2020. arXiv:2003.06988.

The failure modes behind the reachability family, and the Table 1 room-connection counts behind S1, S2 and DR2. Those counts were measured on proximity adjacency.

Hillier, B., & Hanson, J. (1984). *The Social Logic of Space*. Cambridge University Press.

Depth from the entrance, the metric behind DP1, and the genotype of shallow public rooms and deep private ones that PG1 tests for inversion.

SIA 500 — *Hindernisfreie Bauten* (Barrier-free construction). Swiss Society of Engineers and Architects.

The 1.2 m clear circulation width that A1 tests as two grid cells.

SIA 181 — *Schallschutz im Hochbau* (Sound insulation in buildings). Swiss Society of Engineers and Architects.

Separation of noisy uses from quiet ones, grounding AC1.

VKF / AEAI — Swiss Fire Protection Guidelines (Vereinigung Kantonaler Feuerversicherungen).

The escape travel-distance concept behind F1, approximated here with a hop count.

Swiss glazing practice — daylight provision for habitable rooms as a fraction of floor area.

The window targets under D1, D2 and W1: 1/6 for living and recreation, 1/10 for bedrooms and kitchens.

Written from `src/core/rules.ts` and `src/core/windows.ts`; every id, tier and constant here matches the code as built.